FEA OF SPROCKET FOR A FSAE VEHICLE.

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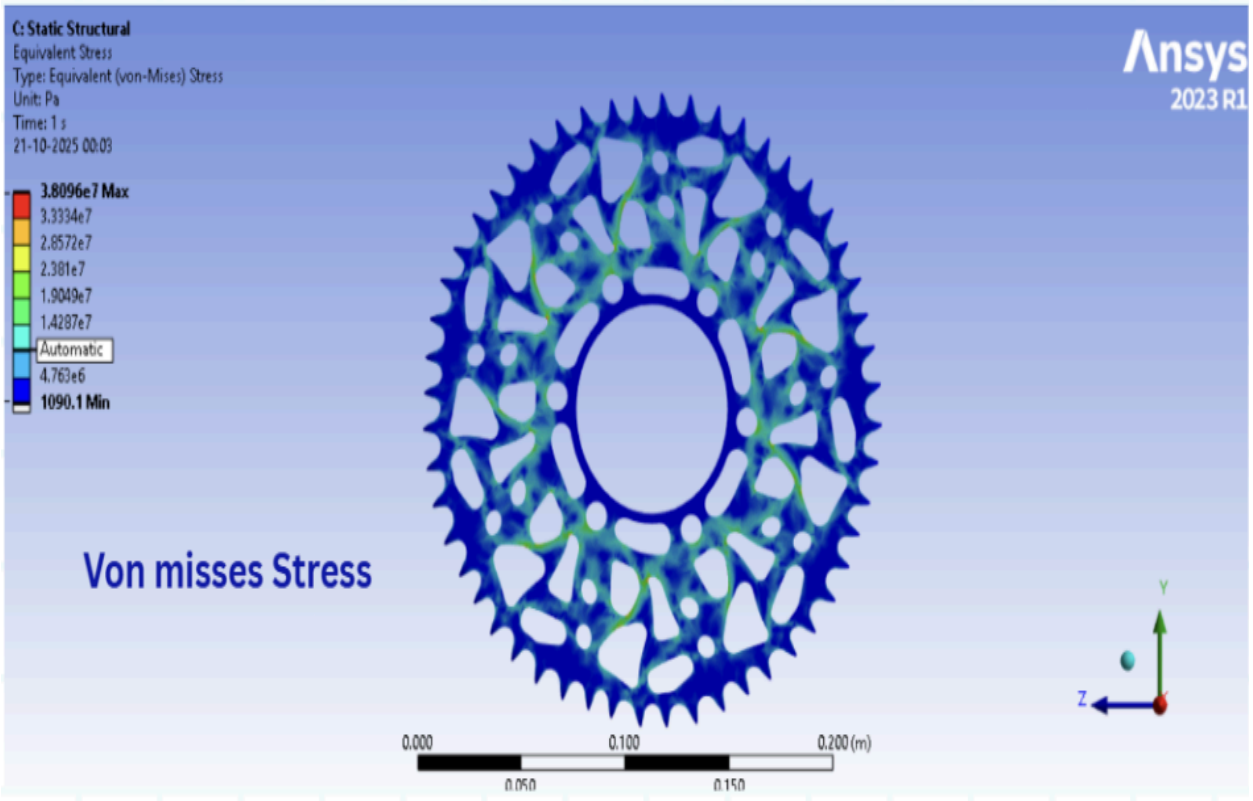
The analysis of a sprocket using ANSYS is a critical step in drivetrain design as it ensures that the component can withstand the high torque loads of the engine while remaining as lightweight as possible to reduce rotational inertia. The maximum Von Mises stress in the sprocket is much lower than the 6061-T6 material's yield strength, resulting in a high Factor of Safety (FOS). Following is the design calculation part :



Sprocket Calculations

The calculations are done for the mechanical sizing and kinematic verification of a chain-drive system. It establishes the physical dimensions and load limits required to ensure the motor's power is effectively transmitted to the wheels. It focuses on external forces and geometry. We have compared the tangential force to the breaking load of the chain (13800N

for a normal 8B chain we used) which is a comparison of ultimate strength.

Motor Specification:

Motor power- 13.4KW

Motor shaft torque- 58Nm

Maximum motor speed- 2200rpm

Driven Sprocket Calculation:

Torque =58*Drive ratio = 58*5.1 = 295.8Nm

No. of teeth of driven gear (Z1) = 52

Pitch circle diameter = P/sin(180/Z1) = 12.7/sin (180/52) =210mm

Diving Sprocket Calculation:

Torque= 58Nm

No. of teeth(Z2) = 52/Drive ratio =52/5.1 ≈11

ISO chain number = 8B

Pitch circle diameter of driving sprocket =P/ sin(180/Z2) =

12.7/sin(180/11) = 45.078mm

Tangential force of chain link at the Driven sprocket:

F=Torque/radius

Radius=210/2 = 105mm

F= 295.8/(105*10^-3) = 2817.14 N

Breaking load of simple 8B chain is 13800N

à Tangential force of the chain link at the Drive Sprocket is significantly lower than the breaking load of a simple 8B chain.

Centre to Centre distance:

C= 121.11mm

Minimum Wrap Angle:

Total no. of engaged teeth = 52-17 = 35

Angle of Rack= (Total no. of engaged teeth * 360)/52 = 242.307

Average Velocity of Chain:

V = (Z2 *P *N)/ (60*10^3) = (11 * 12.7 * 2200) / (60* 10^3)

V == 5.12 m/s

Length of Chain-

L= No. of Links (Ln) * P

Ln= (2C/P) + ((Z1+Z2)/2) + ((Z1-Z2)^2/4πC)

Ln= ((2*121.11)/12.7) + ((52+11)/2) + ((52-11)^2/(4π*121.11))

Ln= 51.68 ≈ 52

L= Ln*P = 52* 12.7 = 660.4 mm